1 Distributional Properties of Categorical Processes

1.1 General Notation

Let (X_t) be a *categorical* time series process, where each X_t takes values in

$$S := \{s_0, \ldots, s_m\}, \qquad m \in \mathbb{N}.$$

(X_t) may be *nominal*, in which case the order of magnitude of the elements in S is arbitrary and solely usesd for notational consistency, or *ordinal*, in which case the order of the elements in S reflects the inherent order of the elements.

Furthermore, (X_t) is assumed to be stationary. A distinction is made between two forms of stationarity. A process is said to be weakly stationary if:

$$\begin{aligned} E(X_t) &= \mu_X \qquad && \forall t, \\ E(X_t^2) &< \infty \qquad && \forall t, \\ \operatorname{Cov}(X_t, X_{t+h}) &= \gamma(h) \qquad && \forall t, h. \end{aligned}$$

In contrast, strong stationarity requires that all finite-dimensional distributions are invariant under time shifts, i.e.,

$$F_{X_t,\ldots,X_{t+j}} = F_{X_{t+h},\ldots,X_{t+h+j}} \quad \forall t, j, h$$

1.2 Marginal Distributions

Regardless of whether we consider a simple categorical process with few categories or attempt to describe a complex, temporally evolving process using moments and measures of serial dependence, the analysis of categorical processes is always built on the examination of the marginal probabilities. However, because ordinal and nominal data encode different types of information, their marginals are typically represented in different ways:

- **Ordinal data:** Most analytical tools for ordinal processes take into account the natural ordering of the elements in S. For this reason, most methods for ordinal data rely on the marginal cumulative distribution function (CDF), which accumulates probabilities in accordance with the natural ordering of the categories:

$$\mathbf{f} = (f_0, \ldots, f_{m-1})^\top, \qquad f_i := P(X_t \leq s_i).$$

For ordinal data, it is sufficient to consider the cumulative probabilities of the categories $0, \dots, m-1$ since the cumulative probability of the last category m is always equal to 1.

Although one could equivalently estimate the PMF and derive the CDF from it, representing ordinal data through the CDF is often more convenient as it this form that is used to compute metrics for location and disperson.

- **Nominal data:** Since the categories of nominal variables do not possess a meaningful order, statistical procedures cannot make use of cumulative structure. Therefore the marginal distribution is best represented by the probability mass function (PMF):

$$\mathbf{p} = (p_0, \dots, p_m)^\top, \qquad p_i := P(X_t = s_i).$$

1.3 Bivariate Probabilities at Lag h

Temporal dependence in categorical time series can be quantified through bivariate lag-h probabilities. These represent the joint probability that the process takes a specific category at time t and another category at time $t+h$

- **Ordinal series (cumulative):**

$$f_{ij}(h) := P(X_t \le s_i,\, X_{t+h} \le s_j), \qquad i, j < m.$$

- **Nominal series (joint):**

$$p_{ij}(h) := P(X_t = s_i,\, X_{t+h} = s_j), \qquad i, j \le m.$$

These quantities serve as input for dependence measures such as the ordinal/ nominal Cohen's K.

1.4 Location

For a nominal process, any measure that relies on ordering is inappropriate. Instead, a location measure must depend only on the category probabilities. The mode, defined as the most probable category,

$$\mathrm{Mod}(X_t) = \arg\max_{s_i \in S} P(X_t = s_i),$$

is therefore the natural choice. It provides a meaningful summary of the process by identifying the most frequent category.

For an ordinal process, the natural ordering of the categories allows for additional measures of central tendency. While the mode can still be informative as the most frequent category, the median respects the ordering of categories and is often more useful for describing the central location:

$$\mathrm{Med}(X_t) = \min\Big\{\, s_i \in S : F(s_i) \geq \tfrac{1}{2} \,\Big\}.$$

Thus, for ordinal data, both the mode and the median carry information, but they capture slightly different aspects of location: the mode identifies the most typical category, whereas the median indicates the midpoint in the ordered scale.

1.5 Measures of Dispersion

The concept of dispersion differs between ordinal and nominal processes. Therefore, two separate measures are used: the *Index of Ordinal Variation* (IOV) for ordinal data and the *Index of Qualitative Variation* (IQV) for nominal data. Both indices are scaled to the interval $[0, 1]$, where a value of 0 represents minimal dispersion and a value of 1 represents maximal dispersion.

$$\mathrm{IOV} = \frac{4}{m}\sum_{i=0}^{m-1} f_i(1-f_i) \qquad \mathrm{IQV} = \frac{m+1}{m}\left(1 - \sum_{i=0}^{m} p_i^2\right)$$

Minimal Dispersion

Both measures attain the value 0 if and only if the distribution is degenerate, that is, if all probability mass is concentrated in a single category.

Ordinal case. A one–point distribution assigns all probability mass to a single category $s_j \in S$. For ordinal data, this corresponds to a CDF vector $\mathbf{f}$ of the form

$$\mathbf{f} \in \left\{ \begin{pmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ \vdots \\ 1 \end{pmatrix}, \dots, \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix} \right\} := \{\boldsymbol{c}_0, \dots, \boldsymbol{c}_m\} \subset [0,1]^{m+1}.$$

In each of these cases, every component f_i satisfies $f_i \in \{0, 1\}$, and thus

$$f_i(1-f_i) = 0 \qquad \forall i.$$

Therefore,

$$\mathrm{IOV} = \frac{4}{m}\sum_{i=0}^{m-1} f_i(1-f_i) = 0.$$

Conversely, if $\mathrm{IOV} = 0$, then

$$\sum_{i=0}^{m-1} f_i(1-f_i) = 0,$$

and since each term is nonnegative,

$$f_i(1-f_i) \overset{!}{=} 0 \quad \forall i.$$

This is only possible if $f_i \in \{0,1\}$ for all i, which implies $\mathbf{f} \in \{\boldsymbol{e}_0, \ldots, \boldsymbol{e}_m\}$. Hence, IOV $= 0$ holds *exactly* for one–point distributions.

Nominal case. A degenerate (one–point) distribution assigns all probability mass to a single category $s_j \in S$, which corresponds to a PMF vector of the form

$$\mathbf{p} \in \left\{ \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix}, \ldots, \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix} \right\} := \{\boldsymbol{e}_0, \ldots, \boldsymbol{e}_m\} \subset [0,1]^{m+1}.$$

In each of these cases, exactly one component equals 1 and all others equal 0, so

$$\sum_{i=0}^{m} p_i^2 = 1.$$

Thus,

$$\text{IQV} = \frac{m+1}{m}\Big(1 - \sum_{i=0}^{m} p_i^2\Big) = \frac{m+1}{m}(1-1) = 0.$$

Conversely, if IQV $= 0$, then

$$\frac{m+1}{m}\left(1 - \sum_{i=0}^{m} p_i^2\right) = 0 \quad \Longrightarrow \quad \sum_{i=0}^{m} p_i^2 = 1.$$

As all probabilities satisfy $0 \le p_i \le 1$, it follows that

$$p_i^2 \le p_i \quad \forall p_i, \text{ and } p_i^2 < p_i \quad \forall p_i \notin \{0,1\}.$$

Since $\sum_{i=1}^{m} p_i = 1$, it follows that the only way $\sum_i p_i^2 = 1$ can hold is if exactly one p_i equals 1 and all others are 0. In other words, the distribution is degenerate and $\boldsymbol{p} \in \{\boldsymbol{e}_0, \ldots, \boldsymbol{e}_m\}$.

Maximal Dispersion

Ordinal case. Dispersion of an ordinal variable is maximal when half of the probability mass lies in the lowest category and half in the highest category; with all intermediate categories having probability zero. This results in a CDF vector,

$$\mathbf{f} = \left(\tfrac{1}{2}, \tfrac{1}{2}, \ldots, \tfrac{1}{2}\right)^{\top}.$$

Then

$$f_i(1-f_i) = \frac{1}{2}\left(1 - \frac{1}{2}\right) = \frac{1}{4} \qquad \forall i,$$

and thus

$$\mathrm{IOV} = \frac{4}{m} \cdot \sum_{i=0}^{m-1} \frac{1}{4} = \frac{4}{m} \cdot \frac{m}{4} = 1.$$

To show that IOV $= 1 \Rightarrow \boldsymbol{f} = (\frac{1}{2}, \ldots, \frac{1}{2})^T$, recall that the IOV only depends on $\boldsymbol{f}$ through the sum

$$\sum_{i=0}^{m-1} f_i(1 - f_i).$$

Taking the derivative of $g(f_i) := f_i(1 - f_i)$ and setting it to zero yields:

$$\begin{aligned}
\frac{\partial}{\partial f_i} g(f_i) &= 1 - 2f_i, \\
1 - 2f_i &\stackrel{!}{=} 0, \\
f_i &= \frac{1}{2}.
\end{aligned}$$

Since $g(f_i = 0) = g(f_i = 1) = 0$, the fucntion is concave and this critical point is indeed the global maximum. Since its been already shown that $IOV(\boldsymbol{f} = (\frac{1}{2}, \ldots, \frac{1}{2})^T) = 1$, it directly follows that

$$IOV(\boldsymbol{f}) = 1 \Leftrightarrow \boldsymbol{f} = (\frac{1}{2}, \ldots, \frac{1}{2})^\top$$

Nominal case. For nominal data, maximal dispersion is attained when probability mass is distributed uniformly across all $m + 1$ categories, i.e.

$$p_i = \frac{1}{m+1}, \qquad i = 0, \ldots, m.$$

In this case,

$$\sum_{i=0}^{m} p_i^2 = (m+1)\frac{1}{(m+1)^2} = \frac{1}{m+1},$$

and hence

$$\mathrm{IQV} = \frac{m+1}{m}\left(1 - \frac{1}{m+1}\right) = 1.$$

Conversely, suppose that IQV $= 1$. Then

$$\frac{m+1}{m}\left(1 - \sum_{i=0}^{m} p_i^2\right) = 1 \quad \Longrightarrow \quad \sum_{i=0}^{m} p_i^2 = \frac{1}{m+1}.$$

Thus, maximizing IQV is equivalent to minimizing

$$f(\boldsymbol{p}) := \sum_{i=0}^{m} p_i^2 \quad \text{subject to} \quad \sum_{i=0}^{m} p_i = 1.$$

Introducing the Lagrangian (with Lagrange multiplier λ)

$$\mathcal{L}(\boldsymbol{p}, \lambda) := \sum_{i=0}^{m} p_i^2 - \lambda\left(\sum_{i=0}^{m} p_i - 1\right),$$

the first-order conditions are

$$\frac{\partial \mathcal{L}}{\partial p_j} = 2p_j - \lambda = 0$$
$$\Rightarrow \quad 2p_j = \lambda, \qquad j = 0, \dots, m,$$

together with the constraint $\sum_{i=0}^{m} p_i = 1$. Since the same multiplier λ applies to all components, it follows that

$$p_0 = p_1 = \cdots = p_m.$$

Imposing the constraint yields

$$(m+1)p_j = 1 \quad \Longrightarrow \quad p_j = \frac{1}{m+1} \quad \forall j \in 0, \dots, m.$$

Hence, $f(\boldsymbol{p})$ is uniquely minimized at

$$\boldsymbol{p} = \left(\tfrac{1}{m+1}, \dots, \tfrac{1}{m+1}\right)^{\top},$$

and therefore IQV attains its maximal value at the uniform distribution.

1.6 Measures of temporal dependence

2 Samples Measures

2.1 Estimation of Marginal Distributions

To estimate the marginal (cumulative) probabilities, we express the relative frequencies through a binarization of the original process (X_t). For ordinal data, define the indicator vector

$$\boldsymbol{Z}_t := (Z_{t,0}, \dots, Z_{t,m-1})^\top, \qquad Z_{t,i} := \mathbb{I}\{X_t \leq s_i\},$$

and for nominal data analogously

$$\boldsymbol{Y}_t := (Y_{t,0}, \dots, Y_{t,m})^\top, \qquad Y_{t,i} := \mathbb{I}\{X_t = s_i\}.$$

The marginal CDF and PMF vectors are then estimated by the corresponding sample means,

$$\hat{\boldsymbol{f}} := \frac{1}{n} \sum_{t=1}^{n} \boldsymbol{Z}_t, \qquad \hat{\boldsymbol{p}} := \frac{1}{n} \sum_{t=1}^{n} \boldsymbol{Y}_t.$$

2.2 Asymptotics of the marginal estimator

To derive the asymptotic distribution of $\hat{\boldsymbol{f}}$ and $\hat{\boldsymbol{p}}$, a central limit theorem for dependent data is required. Since independence is not assumed, we impose weak dependence conditions on the original process (X_t).
Specifically, $(X_t)_{t\in\mathbb{Z}}$ is assumed to be strictly stationary and α-mixing with exponentially decaying coefficients.
Following Bradley (2005), let

$$\mathcal{F}_J^L := \sigma(X_t : J \leq t \leq L)$$

denote the σ-field generated by $\{X_t : J \leq t \leq L\}$. The α-mixing coefficients of (X_t), given that it's strictly stationary, are defined as

$$\alpha_{X_t}(n) := \sup_{A\in\mathcal{F}_{-\infty}^0,\, B\in\mathcal{F}_n^\infty} \Big| P(A\cap B) - P(A)P(B) \Big|.$$

The process is called α-mixing if $\alpha(n) \to 0$ as $n \to \infty$.
We additionally assume exponential decay of the mixing coefficients, i.e., there exist constants $C > 0$ and $\rho \in (0,1)$ such that

$$\alpha(n) \leq C\rho^n \qquad \text{for all } n \in \mathbb{N}.$$

The indicator mappings

$$g_i(X_t) := \mathbb{I}\{X_t \leq s_i\}, \qquad h_i(X_t) := \mathbb{I}\{X_t = s_i\},$$

are measurable functions of X_t (see Klenke, 2005, p. 34). Since measurability preserves strict stationarity, the binarized processes $(g_i(X_t))$ and $(h_i(X_t))$ are strictly stationary whenever (X_t) is strictly stationary (see Kallenberg, 2002, p. 558). Consequently, the vector-valued processes $\boldsymbol{Z}_t$ and $\boldsymbol{Y}_t$ are strictly stationary as well.

Moreover, each indicator variable $Z_{t,i}$ (or $Y_{t,i}$) is defined on the same probability space as X_t, and the σ-field it generates is a sub-σ-field of $\sigma(X_t)$, i.e.

$$\sigma(Z_{t,i}) \subset \sigma(X_t).$$

The same inclusion holds for time-indexed σ-fields, namely

$$\sigma(Z_{t,i} : J \leq t \leq L) \subset \sigma(X_t : J \leq t \leq L), \qquad J \leq L.$$

It therefore follows that, for all $i = 0, \ldots, m$ and all $n \in \mathbb{N}$,

$$\alpha_{Z_{t,i}}(n) \leq \alpha_{X_t}(n),$$

and analogously for $Y_{t,i}$. Hence, the vector-valued processes $\boldsymbol{Z}_t$ and $\boldsymbol{Y}_t$ are also α-mixing.

The central limit theorem applicable under these assumptions is due to Ibragimov (1962, Theorem 7), which requires strict stationarity and α-mixing. In addition, the theorem assumes the summability condition

$$\sum_{n=1}^{\infty} \alpha(n)^{\lambda/(2+\lambda)} < \infty \quad \text{for some } \lambda > 0.$$

This condition is satisfied under exponentially decaying mixing coefficients. Indeed, if $\alpha(n) \leq C\rho^n$ with constants $C > 0$ and $\rho \in (0,1)$, then

$$\begin{aligned}\sum_{n=1}^{\infty} \alpha(n)^{\lambda/(2+\lambda)} &\leq C^{\lambda/(2+\lambda)} \sum_{n=1}^{\infty} \rho^{n\lambda/(2+\lambda)} \\ &= C^{\lambda/(2+\lambda)} \frac{\rho^{\lambda/(2+\lambda)}}{1 - \rho^{\lambda/(2+\lambda)}} < \infty,\end{aligned}$$

since $0 < \rho^{\lambda/(2+\lambda)} < 1$ for all $\lambda > 0$.
Moreover, the CLT additionally requires the existence of moments of order $2 + \lambda$ for some $\lambda > 0$. This condition is trivially satisfied in our setting, since $\boldsymbol{Z}_t$ and $\boldsymbol{Y}_t$ are vectors of indicator variables. In particular, for any

$$|Z_{t,i}|^{2+\lambda} \in \{0, 1\},$$

and hence all higher order moments exist. Consequently, for all $i \in 0, \ldots, m$ and $\lambda > 0$,

$$\mathbb{E}|Z_{t,i}|^{2+\lambda} < \infty,$$

and analogously for $Y_{t,i}$.

According to the Cramér-Wold device, $(\boldsymbol{Z}_t)_t$ converges in distribution to a multivariate normal distribution $\mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ if and only if for every fixed non-zero vector $\boldsymbol{a} \in \mathbb{R}^{m+1} \setminus \boldsymbol{0}$, the scalar projections

$$U_t := \boldsymbol{a}^\top \boldsymbol{Z}_t = \sum_{i=0}^{m} a_i Z_{t,i}$$

converge in distribution to a univariate normal distribution $\mathcal{N}(\boldsymbol{a}^\top \boldsymbol{\mu}, \boldsymbol{a}^\top \boldsymbol{\Sigma} \boldsymbol{a})$.
Let

$$L : \mathbb{R}^{m+1} \to \mathbb{R}, \quad L(z) = \boldsymbol{a}^\top.$$

Since L is linear on a finite dimensional space, it is continuous, and therefore (Borel-) measurable.
The vector-valued function $g(X_t) = \boldsymbol{Z}_t$ is measurable due to the measurability of its components. Therefore, the composition

$$U_t = L \circ g \circ X_t$$

is also measurable.
Since U_t is a measurable function of X_t, it remains its properties of strict stationarity and α-mixing with exponentially decaying coefficients.
Accordingly, the summability condition is satisfied (follows from the α-mixing with exponentially decaying coefficients). Additionally, there will also be a value M such that

$$M \geq \max_{j=0,\dots,m} |\sum_{i=0}^{j} a_i Z_i| < \infty$$

Since $|U_t| \leq M$, we have $|U_t|^{2+\lambda} \leq M^{2+\lambda}$ for every $\lambda > 0$, and therefore

$$E|U_t|^{2+\lambda} < |M|^{2+\lambda} < \infty \quad \forall \lambda > 0$$

Since U_t is strictly stationary, α-mixing and satsifying the summability aswell as the moment condition, the CLT from Ibragimov is applicable. It follows that the sample mean

$$\bar{U}_t := \frac{1}{n} \sum_{t=1}^{n} U_t$$

satisfies the CLT:

$$\sqrt{n}(\bar{U}_t - \mu_{U_t}) \xrightarrow[n\to\infty]{d} \mathcal{N}(0, \sigma_{\boldsymbol{a}}),$$

where

$$\mu_{U_t} = E[U_t] \quad \text{and} \quad \sigma_{\boldsymbol{a}} = Var(U_t) + \sum_{h=1}^{\infty} Cov(U_t, U_{t+h})$$
$$= \sum_{i,j} a_i a_j \left[Cov(Z_{t,i}, Z_{t,j}) + \sum_{h=1}^{\infty} Cov(Z_{t,i}, Z_{t+h,j}) + Cov(Z_{t,j}, Z_{t+h,i}) \right]$$

Hence, according to the Cramer-Wold device, the sample mean

$$\bar{\boldsymbol{Z}}_n := \frac{1}{n}\sum_{t=1}^{n} \boldsymbol{Z}_t = \hat{\boldsymbol{f}}$$

satisfies the following multivariat CLT:

$$\sqrt{n}\Big(\bar{\boldsymbol{Z}}_n - \boldsymbol{f}\Big) \xrightarrow[n\to\infty]{d} \mathcal{N}\Big(\mathbf{0}, \boldsymbol{\Sigma}_Z\Big), \quad \text{with } \boldsymbol{\Sigma}_{\boldsymbol{Z}} = (\sigma_{\boldsymbol{Z};i,j})_{i,j=0,\ldots,m-1}$$

where $\boldsymbol{\Sigma}_Z$ must satisfy that for all $\boldsymbol{a} \in \mathbb{R}^m$

$$\boldsymbol{a^T}\boldsymbol{\Sigma}_{\boldsymbol{Z}}\boldsymbol{a} = \sigma_{\boldsymbol{a}} = \sum_{i,j} a_i a_j \left[Cov(Z_{t,i}, Z_{t,j}) + \sum_{h=1}^{\infty} Cov(Z_{t,i}, Z_{t+h,j}) + Cov(Z_{t,j}, Z_{t+h,i})\right],$$

which implies

$$\sigma_{\boldsymbol{Z};i,j} = Cov(Z_{t,i}, Z_{t,j}) + \sum_{h=1}^{\infty} Cov(Z_{t,i}, Z_{t+h,j}) + Cov(Z_{t,j}, Z_{t+h,i}).$$

Since

$$Cov(Z_{t,i}, Z_{t,j}) = E[Z_{t,i}Z_{t,j}] - E[Z_{t,i}]E[Z_{t,j}] = f_{\min\{i,j\}} - f_i f_j$$

and

$$Cov(Z_{t,i}, Z_{t+h,j}) = E[Z_{t,i}Z_{t+h,j}] - E[Z_{t,i}]E[Z_{t+h,j}] = f_{ij}(h) - f_i f_j$$

we can also write the elements of $\boldsymbol{\Sigma}_{\boldsymbol{Z}}$ as

$$\sigma_{\boldsymbol{Z};i,j} = \Big(f_{\min\{i,j\}} - f_i f_j\Big) + \sum_{h=1}^{\infty} f_{ij}(h) + f_{ji}(h) - 2f_i f_j$$

2.3 Asymptotic Properties of Sample Dispersion Measures

IOV

We can now now use this information of the asymptotic distribution of $\hat{\boldsymbol{f}}$ to compute the asymptotic properties of sample measures that are based on $\hat{\boldsymbol{f}}$. According to the multivariate delta-method a random variable $G^* := g^*(\boldsymbol{\Theta^*})$, where $\boldsymbol{\Theta^*} \in \mathbb{R}^m$, $\boldsymbol{\Theta^*} \overset{asy.}{\sim} \mathcal{N}(\boldsymbol{\Theta}, \boldsymbol{n^{-1}}\boldsymbol{\Sigma}^*)$ and $g : \mathbb{R}^m \to \mathbb{R}$ is a continously differentiable function, satisfies

$$G^* \overset{asy.}{\sim} \mathcal{N}(g^*(\boldsymbol{\Theta}), n^{-1}\boldsymbol{j}\boldsymbol{\Sigma}^*\boldsymbol{j}^\top)$$

where $\boldsymbol{j}$ is the one-dimensional Jacobian-matrix (i.e. vector) and where $j_i = \frac{\partial g^*(\boldsymbol{\Theta})}{\partial \theta_i}$. Since we have shown that $\hat{\boldsymbol{f}}$ is assymptotically normal we can use this to derive the asymptotic properties of sample measures like the IOV. The respective Jacobian is given by:

$$\boldsymbol{j}_{IOV} = \left(\frac{\partial IOV(\boldsymbol{f})}{\partial f_0}, \ldots, \frac{\partial IOV(\boldsymbol{f})}{\partial f_{m-1}}\right) = \frac{4}{m}\left((1-2f_0), \ldots, (1-2f_{m-1})\right)$$

which leads to an asymptotic variance of the IOV:

$$\begin{aligned} Var(\widehat{IOV(\boldsymbol{f})}) &\overset{asy.}{=} \boldsymbol{j}_{IOV}\boldsymbol{\Sigma}_Z\boldsymbol{j}_{IOV}^\top = \sum_{i=0}^{m-1}\sum_{j=0}^{m-1}\frac{4(1-2f_i)}{m}\frac{4(1-2f_j)}{m}\sigma_{\boldsymbol{Z};i,j} \\ &= \frac{16}{m^2}\sum_{i,j=0}^{m-1}(1-2f_i)(1-2f_j)\sigma_{\boldsymbol{Z};i,j} \end{aligned}$$

However, since IOV is not a linear function, the first-order delta method only provides information about the asymptotic distribution and variance, not on the finite-sample bias:

$$(\widehat{IOV} \xrightarrow{p} IOV(\boldsymbol{f})) \not\Rightarrow (E[\widehat{IOV}] = IOV(\boldsymbol{f})).$$

For non linear transformations, the curvature induces a bias of order n^{-1}. Since the IOV-function is continouosly differentiable twice, we can quantify this additional second-order effect by expanding $IOV(\hat{\boldsymbol{f}})$ in a Taylor series up to second order:

$$IOV(\hat{\boldsymbol{f}}) = IOV(\boldsymbol{f}) + \boldsymbol{j}_{IOV}(\hat{\boldsymbol{f}} - \boldsymbol{f}) + \frac{1}{2}(\hat{\boldsymbol{f}} - \boldsymbol{f})^\top \boldsymbol{H}_{IOV}(\boldsymbol{f})(\hat{\boldsymbol{f}} - \boldsymbol{f}) + o_p(n^{-1}),$$

where $\boldsymbol{H}_{IOV}(\boldsymbol{f})$ is the Hessian matrix such that

$$\boldsymbol{H}_{IOV}(\boldsymbol{f}) = \left(\frac{\partial^2 IOV(\boldsymbol{f})}{\partial f_i f_j}\right)_{i,j=0,\ldots,m-1} = \begin{cases} -\frac{8}{m} & \text{if } i = j \\ 0 & \text{else} \end{cases} = -\frac{8}{m}\boldsymbol{I}_m$$

Since

$$E(\boldsymbol{j}_{IOV}(\hat{\boldsymbol{f}} - \boldsymbol{f})) = 0,$$

the finite sample bias is given by

$$\begin{aligned} Bias(IOV(\hat{\boldsymbol{f}})) &= E[\frac{1}{2}(\hat{\boldsymbol{f}} - \boldsymbol{f})^\top \boldsymbol{H}_{IOV}(\boldsymbol{f})(\hat{\boldsymbol{f}} - \boldsymbol{f})] = \frac{1}{2}tr(\boldsymbol{H}_{IOV}Cov(\hat{\boldsymbol{f}})) \\ &= -\frac{4}{m}\sum_{i=0}^{m-1}\sigma_{\boldsymbol{Z};i,i}. \end{aligned}$$

This implies that the bias is strictly negative and only depends on the marginal variances $\sigma_{\boldsymbol{Z};i,i}$ of the indivdual estimators f_i, not on the covariances.

Skew

In a similiar way the asymptotic properties of other dispersion metrics such as the skew can be analyzed. A simple measure for the skew of an ordinal process is given by:

$$\text{skew}(\boldsymbol{f}) = \frac{2}{m} \sum_{i=0}^{m-1} f_i - 1.$$

The jacobian for the skew is given by

$$\boldsymbol{j}_{skew} = \left(\frac{\partial skew(\boldsymbol{f})}{\partial f_0}, \ldots, \frac{\partial skew(\boldsymbol{f})}{\partial f_{m-1}} \right) = \left(\frac{2}{m}, \ldots, \frac{2}{m} \right)$$

such that we end up with assymptotic expectation and variance:

$$\begin{aligned} Var(\widehat{\text{skew}(\boldsymbol{f})}) &\overset{asy.}{=} n^{-1} \boldsymbol{j}_{skew} \Sigma_{\boldsymbol{Z}} \boldsymbol{j}_{skew}^\top = \frac{1}{n} \frac{4}{m^2} \sum_{i,j=0}^{m-1} \sigma_{\boldsymbol{Z};ij} \\ E(\widehat{\text{skew}(\boldsymbol{f})}) &\overset{asy.}{=} \text{skew}(E(\hat{\boldsymbol{f}})) = \text{skew}(\boldsymbol{f}). \end{aligned}$$

Since the skew is linear in $\boldsymbol{f}$, the Hessian vanishes ($\boldsymbol{H}_{skew} = 0$). It follows that the estimator is exactly unbiased for all sample sizes n, not only asymptotically.
The derivations for other dispersion metrics are analogous.

3 Introducing Missingness

To model the missingness, the concept of **amplitude modulation** is employed. This approach models missingness through a binary stochastic process $(O_t)_t$ that runs concurrently with the main process $(X_t)_t$ and modulates its "amplitude" in a binary manner:

$$O_t \in \{0, 1\} \quad \forall t$$

where $O_t = 1$ indicates that X_t is fully observed (amplitude = 1), and $O_t = 0$ indicates that X_t is missing (amplitude = 0).

The amplitude modulating process O_t can be:

- **Deterministic**: For each t, $O_t = o_t$ where $o_t \in \{0, 1\}$ is a known, fixed value.
- **Stochastic**: O_t follows a probability distribution with $\pi_t := P(O_t = 1)$, possibly dependent on the history of X_t, O_t, or external factors.

The amplitude modulation is applied to the binarization vector $\boldsymbol{Z}_t$ to obtain the observable process $(O_t \boldsymbol{Z}_t)$. We then define the naive estimator of $\boldsymbol{f}$ as:

$$\hat{\mathfrak{f}} := \frac{1}{n} \sum_{t=1}^{n} O_t \boldsymbol{Z}_t.$$

To assess the performance of this naive estimator we compute its expectation

$$E[\hat{\mathfrak{f}}] = \frac{1}{n} \sum_{t=1}^{n} E[O_t \boldsymbol{Z_t}] = \frac{1}{n} \sum_{t=1}^{n} E[\boldsymbol{Z_t} E[O_t | \boldsymbol{Z}_t]],$$

where the last equality follows from the law of iterated expectations.
The dependendce structure between $(O_t)_t$ and $(\boldsymbol{Z}_t)_t$ (or equivalently $(X_t)_t$) is crucial. Under the assumption of independence, the expression simplifies to

$$\frac{1}{n} \sum_{t=1}^{n} E[\boldsymbol{Z_t} E[O_t | \boldsymbol{Z}_t]] = \frac{1}{n} \sum_{t=1}^{n} E[\boldsymbol{Z}_t] E[O_t] = \left(\frac{1}{n} \sum_{t=1}^{n} E[O_t] \right) \boldsymbol{f}$$

Since $E[\hat{\mathfrak{f}}]$ deviates from $\boldsymbol{f}$ by the factor $\frac{1}{n} \sum_{t=1}^{n} E[O_t]$, it implies that $\boldsymbol{f}$ should be estimated using the corrected estimator:

$$\hat{\boldsymbol{f}}^* := \frac{\frac{1}{n} \sum_{t=1}^{n} O_t \boldsymbol{Z}_t}{\frac{1}{n} \sum_{t=1}^{n} O_t} =: \frac{\hat{\mathfrak{f}}}{\bar{O}}$$

Under the assumption of the independence of (O_t) and (X_t), the stochastic properties of the corrected estimator $\hat{\boldsymbol{f}}^*$ can be studied. If (O_t) is deterministic, then $E[O_t] = o_t \in \{0, 1\}$ for any given t. It follows that

$$\frac{1}{n}\sum_{t=1}^{n} E[O_t] = \frac{1}{n}\sum_{t=1}^{n} O_t = \bar{O},$$

and therefore, here $\hat{\boldsymbol{f}}^* = \hat{\mathfrak{f}}/\bar{O}$ is a perfectly unbiased estimator of $\boldsymbol{f}$. However, if O_t is a stochastic process then $E[O_t] = \pi_t \in (0; 1]$ and

$$\frac{1}{n}\sum_{t=1}^{n} E[O_t] = \frac{1}{n}\sum_{t=1}^{n} \pi_t \neq \frac{1}{n}\sum_{t=1}^{n} O_t = \bar{O},$$

to where $\hat{\boldsymbol{f}}^*$ is generally biased. To study this bias we assume (O_t) and (X_t) to be stationary and α-mixing with exponetially decaying coefficients, with $E[O_t] = \pi$ and a autocovariance function $\gamma_O(h) := Cov[O_h, O_0]$ and joint expectations $\pi(h) := E[O_h O_0] = \gamma_O(h) + \pi^2$. To study the joint asymptotic distribution of the estimator $\hat{\boldsymbol{f}}^*$, we introduce the extended binary vectors in order to derive the joint asymptotic distribution of all sample averages entering its definition:

$$\boldsymbol{Z}_t^* := (O_t, O_t Z_{t,0}, \ldots, O_t Z_{t,0})^\top = O_t(1, \boldsymbol{Z}_t^\top)^\top,$$

where the elements of $\boldsymbol{Z}_t^*$ are are numbered as $Z_{t,-1}^*, Z_{t,0}^*, \ldots, Z_{t,m-1}^*$The expectation of $\boldsymbol{Z}_t^*$ is given by

$$\begin{aligned}
E[\boldsymbol{Z}_t^*] &= (E[O_t], E[O_t Z_{t,0}], \ldots, E[O_t Z_{t,m-1}])^\top \\
&= (E[O_t], E[O_t]E[Z_{t,0}], \ldots, E[O_t]E[Z_{t,m-1}])^\top && \text{(Independence)} \\
&= (\pi, \pi f_0, \ldots, \pi f_{m-1})^\top \\
&= \pi(1, \boldsymbol{f}^\top)^\top =: \boldsymbol{\mu}_{\boldsymbol{Z}_t^*}.
\end{aligned}$$

Since the processes (O_t) and $(\boldsymbol{Z}_t)$ are independent and stationary, the joint process $((O_t, \boldsymbol{Z}_t)^\top)$ is also stationary. Moreover, if both processes are α-mixing, then the joint process is α-mixing. As $\boldsymbol{Z}_t^*$ is a measurable function of $(O_t, \boldsymbol{Z})^\top$ it follows that $\boldsymbol{Z}_t^*$ is stationary and α-mixing (with exponentially decaying coefficients).
To study the asymptotic properties of $\boldsymbol{Z}_t^*$ we once again use Cramer-Wold. Let

$$U_t^* = \boldsymbol{a}^\top \boldsymbol{Z}_t^*, \quad \boldsymbol{a} \in \mathbb{R}^{m+1}.$$

The argument for why U_t* satisfies the conditions of Ibragimovs CLT is exactly the same as in the argument for why the linear projections of the non-missing estimator $\boldsymbol{Z}_t$ (U_t) converges. It follows that

$$\sqrt{n}(\bar{U}_t^* - \mu_{U_t^*}) \xrightarrow[n\to\infty]{d} \mathcal{N}(0, \sigma_a^*)$$

where $\mu_{U_t^*} = E(U_t) = \boldsymbol{a}^\top \boldsymbol{\mu}_{\boldsymbol{Z}_t^*}$ and

$$
\begin{aligned}
\sigma_a^* &= Var(U_t^*) + 2\sum_{h=1}^{\infty} Cov(U_t, U_{t+h}) \\
\text{where} \quad Var(U_t^*) &= \boldsymbol{a}^\top Var(\boldsymbol{Z}_t^*)\boldsymbol{a} = \sum_{i,j=-1}^{m-1} a_{i+1}a_{j+1} Cov(Z_{t,i}^*, Z_{t,j}^*) \\
\text{and} \quad Cov(U_t^*, U_{t+h}^*) &= \boldsymbol{a}^\top Cov(\boldsymbol{Z}_t^*, \boldsymbol{Z}_{t+h}^{*\top})\boldsymbol{a} \\
\text{such that} \quad \sigma_a^* &= \boldsymbol{a}^\top (Var(\boldsymbol{Z}_t^*) + 2\sum_{h=1}^{\infty} Cov(\boldsymbol{Z}_t^*, \boldsymbol{Z}_{t+h}^*))\boldsymbol{a}.
\end{aligned}
$$

According to Cramer-Wold we can follow, that since every linear projection U_t^* in asymptotitcally normal we can follow that the sample mean

$$\bar{\boldsymbol{Z}}_n^* := \frac{1}{n}\sum_{t=1}^{n} \boldsymbol{Z}_t^*$$

satisfies the multivariat CLT

$$\sqrt{n}\left(\bar{\boldsymbol{Z}}_n^* - \boldsymbol{\mu}_{\boldsymbol{Z}_t^*}\right) \xrightarrow[n\to\infty]{d} \mathcal{N}\left(\boldsymbol{0}, \boldsymbol{\Sigma}_{\boldsymbol{Z}^*}\right)$$

where $\boldsymbol{\Sigma}_{\boldsymbol{Z}^*}$ must sastisfy for all $a \in \mathbb{R}^{m+1}$

$$\boldsymbol{a}^\top \boldsymbol{\Sigma}_{\boldsymbol{Z}^*}\boldsymbol{a} = \sigma_a^* = \boldsymbol{a}^\top (Var(\boldsymbol{Z}_t^*) + 2\sum_{h=1}^{\infty} Cov(\boldsymbol{Z}_t^*, \boldsymbol{Z}_{t+h}^*))\boldsymbol{a}$$

which indicates

$$\boldsymbol{\Sigma}_{\boldsymbol{Z}^*} = Var(\boldsymbol{Z}_t^*) + 2\sum_{h=1}^{\infty} Cov(\boldsymbol{Z}_t^*, \boldsymbol{Z}_{t+h}^*).$$

The individudal elements of $\boldsymbol{\Sigma}_{\boldsymbol{Z}^*}$ are given by

$$\sigma_{\boldsymbol{Z}^*;i,j} = \begin{cases} \text{(i) } i=j=-1: \\ Var(O_t) + 2\sum_{h=1}^{\infty} Cov(O_t, O_{t+h}) \\ \quad = \pi(1-\pi) + 2\sum_{h=1}^{\infty}\gamma_O(h) \\ \text{(ii) } j > i = -1: \\ Cov(O_t, O_t Z_{t,j}) + 2\sum_{h=1}^{\infty} Cov(O_t, O_{t+h}Z_{t+h,j}) \\ \quad = (E[O_t^2]E[Z_{t,j}] - E[O_t]E[O_t]E[Z_{t,j}]) \\ \quad + 2\sum_{h=1}^{\infty}(E[O_tO_{t+h}]E[Z_{t,j}] - E[O_t]E[O_{t+h}]E[Z_{t,j}]) \\ \quad = (\pi(1-\pi)+\pi^2)f_j - \pi^2 f_j + 2\sum_{h=1}^{\infty}(\gamma_O(h)+\pi^2)f_j - \pi^2 f_j \\ \quad = f_j(\pi(1-\pi) + 2\sum_{h=1}^{\infty}\gamma_O(h)) = f_j\sigma_{\boldsymbol{Z}*;-1,-1} \\ \text{(iii) } i,j \geq 0: \\ Cov(O_tZ_{t,j}, O_tZ_{t,i}) + \sum_{h=1}^{\infty} Cov(O_tZ_{t,j}, O_{t+h}Z_{t+h,i}) + Cov(O_{t+h}Z_{t+h,j}, O_tZ_{t,i}) \\ \quad = (E[O_t^2]E[Z_{t,i}Z_{t,j}] - \pi^2 f_i f_j) \\ \quad + \sum_{h=1}^{\infty}(E[O_tO_{t+h}]E[Z_{t,j}Z_{t+h,i}] - \pi^2 f_i f_j) + (E[O_tO_{t+h}]E[Z_{t+h,j}Z_{t,i}] - \pi^2 f_i f_j) \\ \quad = \pi f_{\min\{i,j\}} - \pi^2 f_i f_j + \sum_{h=1}^{\infty}\pi(h)(f_{ij}(h) + f_{ji}(h)) - 2\pi^2 f_i f_j \\ \quad = \pi f_{\min\{i,j\}} - \pi f_i f_j - \pi(1-\pi) f_i f_j \\ \quad + \sum_{h=1}^{\infty}\pi(h)(f_{ij}(h)+f_{ji}(h)) - 2\pi(h) f_i f_j + 2(\pi(h)-\pi^2) f_i f_j \\ \quad = f_i f_j \pi + \pi(f_{\min\{i,j\}} - f_i f_j) + \sum_{h=1}^{\infty}\pi(h)(f_{ij}(h)+f_{ji}(h) - 2f_if_j) + 2(\pi(h)-\pi^2)f_if_j \\ \quad = f_i f_j \sigma_{\boldsymbol{Z}^*;-1,-1} + \pi(f_{\min\{i,j\}} - f_i f_j) + \sum_{h=1}^{\infty}\pi(h)(f_{ij}(h)+f_{ji}(h) - 2f_if_j) \end{cases}$$

To now derive the asymptotic properties of $\hat{\boldsymbol{f}}^*$ we define the function $\boldsymbol{g} := [0,1]^{m+1} \to [0,1]^m$ as

$$g_j(x_{-1}, x_0, \ldots, x_{m-1}) = x_j/x_{-1}, \quad \text{for } j = 0\ldots, m-1,$$

such that

$$g(\bar{\boldsymbol{Z}}_t^*) = \hat{\boldsymbol{f}}^* \text{ and } g(\bar{\boldsymbol{\mu}}_{\boldsymbol{Z}^*}^*) = \boldsymbol{f}.$$

The Jacobian matrix of $\boldsymbol{g}(\boldsymbol{x})$ evaluated at $\boldsymbol{x} = \boldsymbol{\mu}_{\boldsymbol{Z}^*}$ is given by

$$\boldsymbol{G}_{(m+1)\times m} = \begin{pmatrix} \frac{\partial g_0(\boldsymbol{x})}{\partial Z^*_{-1}} & \dots & \frac{\partial g_0(\boldsymbol{x})}{\partial Z^*_{m-1}} \\ \vdots & \dots & \vdots \\ \frac{\partial g_{m-1}(\boldsymbol{x})}{\partial Z^*_{-1}} & \dots & \frac{\partial g_{m-1}(\boldsymbol{x})}{\partial Z^*_{m-1}} \end{pmatrix} = \begin{pmatrix} -\frac{f_0}{\pi} & \frac{1}{\pi} & 0 & \dots & 0 \\ -\frac{f_1}{\pi} & 0 & \frac{1}{\pi} & \ddots & \vdots \\ \vdots & \vdots & \ddots & \ddots & 0 \\ -\frac{f_{m-1}}{\pi} & 0 & \dots & 0 & \frac{1}{\pi} \end{pmatrix}$$

Then, following the deltamethod, the following CLT holds

$$\sqrt{n}(\hat{\boldsymbol{f}}^* - \boldsymbol{f}) \xrightarrow[n\to\infty]{d} \mathcal{N}(\boldsymbol{0}, \boldsymbol{\Sigma}_{\hat{\boldsymbol{f}}^*}),$$

where

$$\begin{aligned}
\boldsymbol{\Sigma}_{\hat{\boldsymbol{f}}^*} &= \boldsymbol{G}\boldsymbol{\Sigma}_{\boldsymbol{Z}^*}\boldsymbol{G}^\top \\
&= \begin{pmatrix} -\frac{f_0}{\pi} & \frac{1}{\pi} & 0 & \dots & 0 \\ -\frac{f_1}{\pi} & 0 & \frac{1}{\pi} & \ddots & \vdots \\ \vdots & \vdots & \ddots & \ddots & 0 \\ -\frac{f_{m-1}}{\pi} & 0 & \dots & 0 & \frac{1}{\pi} \end{pmatrix} \begin{pmatrix} \sigma_{\boldsymbol{Z}^*;-1,-1} & \dots & \sigma_{\boldsymbol{Z}^*;-1,m-1} \\ \vdots & \ddots & \vdots \\ \sigma_{\boldsymbol{Z}^*;m-1,-1} & \dots & \sigma_{\boldsymbol{Z}^*;m-1,m-1} \end{pmatrix} \boldsymbol{G}^\top \\
&= \begin{pmatrix} -\frac{f_0}{\pi}\sigma_{\boldsymbol{Z}^*;-1,-1} + \frac{1}{\pi}\sigma_{\boldsymbol{Z}^*;0,-1} & \dots & -\frac{f_0}{\pi}\sigma_{\boldsymbol{Z}^*;-1,m-1} + \frac{1}{\pi}\sigma_{\boldsymbol{Z}^*;0,m-1} \\ \vdots & \ddots & \vdots \\ -\frac{f_{m-1}}{\pi}\sigma_{\boldsymbol{Z}^*;-1,-1} + \frac{1}{\pi}\sigma_{\boldsymbol{Z}^*;m-1,-1} & \dots & -\frac{f_{m-1}}{\pi}\sigma_{\boldsymbol{Z}^*;-1,m-1} + \frac{1}{\pi}\sigma_{\boldsymbol{Z}^*;m-1,m-1} \end{pmatrix} \boldsymbol{G}^\top \\
\Rightarrow \sigma_{\hat{\boldsymbol{f}}^*;i,j} &= (-\frac{f_j}{\pi})(-\frac{f_i}{\pi}\sigma_{\boldsymbol{Z}^*;-1,-1} + \frac{1}{\pi}\sigma_{\boldsymbol{Z}^*;i,-1}) + \frac{1}{\pi}(-\frac{f_i}{\pi}\sigma_{\boldsymbol{Z}^*;-1,j} + \frac{1}{\pi}\sigma_{\boldsymbol{Z}^*;i,j}) \\
&= \frac{f_j f_i}{\pi^2}\sigma_{\boldsymbol{Z}^*;-1,-1} - \frac{f_j}{\pi^2}\sigma_{\boldsymbol{Z}^*;i,-1} - \frac{f_i}{\pi^2}\sigma_{\boldsymbol{Z}^*;-1,j} + \frac{1}{\pi^2}\sigma_{\boldsymbol{Z}^*;i,j} \\
&= \frac{1}{\pi^2}\left(f_i f_j \sigma_{\boldsymbol{Z}^*;-1,-1} - f_i f_j \sigma_{\boldsymbol{Z}^*;-1,-1} - f_i f_j \sigma_{\boldsymbol{Z}^*;-1,-1} + \sigma_{\boldsymbol{Z}^*;i,j}\right) = \frac{1}{\pi^2}(\sigma_{\boldsymbol{Z}^*;i,j} - f_i f_j \sigma_{\boldsymbol{Z}^*;-1,-1}) \\
&= \frac{1}{\pi^2}\left(\left[f_i f_j \sigma_{\boldsymbol{Z}^*;-1,-1} + \pi(f_{\min\{i,j\}} - f_i f_j) + \sum_{h=1}^{\infty}\pi(h) f_{ij}(h) + f_{ji}(h) - 2 f_i f_j\right] - f_i f_j \sigma_{\boldsymbol{Z}^*;-1,-1}\right) \\
&= \frac{1}{\pi}(f_{\min\{i,j\}} - f_i f_j) + \frac{1}{\pi^2}\sum_{h=1}^{\infty}\pi(h)(f_{ij}(h) + f_{ji}(h) - 2 f_i f_j)
\end{aligned}$$

Since g_j is not a linear function of $\boldsymbol{Z}^*_t$, the bias of order $O(n^{-1})$ is computable using a second-order Taylor expansion. The Hessian of g_j evaluated in $\boldsymbol{\mu}_{\boldsymbol{Z}^*}$ is given by

$$\boldsymbol{H}_{g_j} = \left(\frac{\partial^2 g_j(\boldsymbol{Z}^*_t)}{\partial x_i \partial x_k}\right)_{i,k=-1,\dots,m-1} = \begin{cases} \frac{2f_j}{\pi^2} & \text{if } i = k = -1 \\ -\frac{1}{\pi^2} & \text{if } i = -1,\ k = j \\ 0 & \text{else.} \end{cases}$$

Since only two entries of $\boldsymbol{H}_{g_j}$ are nonzero, and using the symmetry $\sigma_{\boldsymbol{Z}^*;-1,j} = \sigma_{\boldsymbol{Z}^*;j,-1}$, the bias reduces to

$$
\begin{aligned}
\mathrm{Bias}(\hat{f}_j) &= \frac{1}{2}\mathrm{tr}(\boldsymbol{H}_{g_j}\boldsymbol{\Sigma}_{\boldsymbol{Z}^*}) \\
&= \frac{1}{2}h_{g_j;-1,-1}\,\sigma_{\boldsymbol{Z}^*;-1,-1} + h_{g_j;-1,j}\,\sigma_{\boldsymbol{Z}^*;-1,j} \\
&= \frac{1}{2}\cdot\frac{2f_j}{\pi^2}\,\sigma_{\boldsymbol{Z}^*;-1,-1} - \frac{1}{\pi^2}\cdot f_j\,\sigma_{\boldsymbol{Z}^*;-1,-1} \\
&= \frac{f_j}{\pi^2}\,\sigma_{\boldsymbol{Z}^*;-1,-1} - \frac{f_j}{\pi^2}\,\sigma_{\boldsymbol{Z}^*;-1,-1} = 0,
\end{aligned}
$$

where the last step uses $\sigma_{\boldsymbol{Z}^*;-1,j} = f_j\,\sigma_{\boldsymbol{Z}^*;-1,-1}$ from case (ii) of $\boldsymbol{\Sigma}_{\boldsymbol{Z}^*}$.
To further study the asymtpotic distribution of $\hat{\boldsymbol{f}}^*$, we look at a couple more specifc scenarios. For example if the process is fully observed, such that $\pi = 1$. Then

$$
\sigma_{\hat{\boldsymbol{f}}^*;i,j} = (f_{\min\{i,j\}} - f_i f_j) + \sum_{h=1}^{\infty} f_{ij}(h) + f_{ji}(h) - 2f_i f_j = \sigma_{\boldsymbol{Z};i,j}.
$$

If O_t is i.i.d., then $\pi(h) = \pi^2$ such that

$$
\sigma_{\hat{\boldsymbol{f}}^*;i,j} = \frac{1}{\pi}(f_{\min\{i,j\}} - f_i f_j) + \sum_{h=1}^{\infty} f_{ij}(h) + f_{ji}(h) - 2f_i f_j.
$$

By contrast, if X_t is i.i.d., then $f_{ij}(h) = f_i f_j$ for all $i, j = 0, \ldots, m-1$ and

$$
\sigma_{\hat{\boldsymbol{f}}^*;i,j} = \frac{1}{\pi}(f_{\min\{i,j\}} - f_i f_j) + \frac{1}{\pi^2}\sum_{h=1}^{\infty} \pi(h)\overbrace{(f_{ij}(h) + f_{ji}(h) - 2f_i f_j)}^{=0} = \frac{1}{\pi}(f_{\min\{i,j\}} - f_i f_j).
$$

Asymptotic marginal properties

In the same way as for $\hat{\boldsymbol{f}}$ wen can now use the delta method, to learn more about the asymptotic distribution of marginal metrics that are based on $\hat{\boldsymbol{f}}^*$ such as the IOV.